

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 Assessment Tool 2 – Architecture Design Assignment

Course Code:	CSA10	Duration:	60 minutes
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CO Assessed:	CO2	Weightage:	20%
Assessment:	Assessment Tool 2 – Architecture Design Assignment	Problem No.:	15
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Problem Statement 15

Smart Classroom Attendance, Engagement, and Performance Analytics System

Architecture Design Analysis Report

Industry Context

Educational institutions worldwide are rapidly adopting digital technologies to strengthen classroom management and improve student learning outcomes. Advances in artificial intelligence, facial recognition, learning analytics, and cloud-based education platforms now make it possible to automate attendance tracking and to analyse student engagement and academic performance in real time. These capabilities are shifting classroom management away from manual, time-consuming processes toward automated, data-driven, and predictive models that support both administrators and instructors.

Problem Statement

Traditional attendance systems and manual performance evaluation consume valuable instructional time and provide limited insight into student engagement. Teachers often find it difficult to identify at-risk students early enough to intervene effectively. A smart classroom analytics platform is therefore required that can automatically record attendance using AI-based facial recognition or RFID, analyse classroom participation, monitor academic performance continuously, and generate predictive insights for instructors. The system should help educators implement personalised interventions and improve overall learning effectiveness.

Objectives

1. Automate student attendance recording.
2. Analyse classroom engagement.
3. Monitor academic performance continuously.
4. Predict at-risk students using AI.
5. Generate learning analytics dashboards.
6. Support personalised learning strategies.
7. Reduce administrative workload.
8. Improve teaching effectiveness.

Expected Outcomes

1. Accurate attendance management.
2. Improved student engagement.
3. Early identification of academically at-risk students.
4. Better learning outcomes.
5. Reduced manual administrative effort.
6. Data-driven teaching decisions.
7. Enhanced academic performance monitoring.
8. Increased institutional efficiency.

1. Analyze System Requirements

1.1 System Objectives

Based on the assigned problem statement, the primary objective of the system is to eliminate manual, time-consuming attendance and performance-tracking processes by automatically capturing attendance through AI-based facial recognition or RFID, continuously analysing student engagement and academic performance, and using predictive analytics to flag at-risk students so that instructors can intervene early and personalise their teaching strategies.

1.2 Functional Requirements

- The system shall automatically capture and record student attendance using facial-recognition cameras or RFID card readers installed in classrooms.
- The system shall verify student identity against an enrolled student database and log attendance with timestamp and classroom/session details.
- The system shall continuously monitor classroom engagement indicators (e.g., participation frequency, response activity, attentiveness cues) during a session.
- The system shall integrate with assessment, assignment, and examination data sources to track academic performance over time.
- The system shall apply machine-learning models to identify students who are academically at risk based on attendance, engagement, and performance trends.
- The system shall generate real-time and historical analytics dashboards for teachers, department heads, and administrators.
- The system shall send automated notifications/alerts to instructors and, where configured, to parents or mentors for at-risk students.
- The system shall support role-based access so that teachers, students, parents, and administrators view only the data relevant to their role.
- The system shall allow administrators to generate consolidated attendance and performance reports for accreditation and compliance purposes.

1.3 Non-Functional Requirements

- **Scalability:** the system must support many classrooms, thousands of students, and concurrent video/RFID data streams across a campus or multiple campuses without performance degradation.
- **Performance:** facial recognition and attendance logging must complete within a few seconds per student so that classroom time is not disrupted.
- **Reliability:** attendance and performance records must be captured accurately and consistently, with no data loss during peak class-changeover periods.
- **Availability:** the platform must remain available during all instructional hours, with redundancy to avoid disruption to attendance capture or dashboard access.
- **Security & Privacy:** biometric data, academic records, and personal information must be encrypted, access-controlled, and handled in compliance with data-protection regulations.
- **Maintainability:** individual modules (attendance, analytics, prediction, notification) must be independently upgradable, e.g., to retrain ML models without redeploying the whole system.

- **Usability:** dashboards must present complex analytics in an intuitive, actionable format for non-technical teaching staff.
- **Interoperability:** the system must integrate cleanly with existing LMS, ERP, and institutional identity-management systems.

1.4 Key Quality Attributes Influencing the Architecture

The requirements above indicate that the architecture must primarily satisfy scalability (to support campus-wide concurrent AI processing), performance (near real-time recognition and analytics), reliability and availability (uninterrupted attendance capture), security (protection of biometric and academic data), and maintainability (independent evolution of AI models and services). These quality attributes directly shape the architectural style selected in the next section.

2. Select a Suitable Software Architecture

2.1 Selected Architectural Style

A Hybrid Architecture combining Microservices Architecture with an Event-Driven backbone is selected for the Smart Classroom Attendance, Engagement, and Performance Analytics System. Independent, loosely coupled microservices (Attendance, Engagement Analytics, Performance Monitoring, Predictive Analytics, Notification, and User Management) are exposed through a central API Gateway, while an event bus / message broker (e.g., Kafka) handles the high-volume, real-time streams generated by classroom cameras, RFID readers, and IoT sensors.

2.2 Justification for the Selected Architecture

- **Independent scalability:** AI-intensive services such as facial recognition and predictive analytics can be scaled independently of lighter services like notifications, which is essential given the computational load of computer-vision processing across many classrooms.
- **Real-time processing:** the event-driven backbone allows attendance events, engagement signals, and sensor data to be streamed and processed asynchronously, ensuring attendance capture is not blocked by downstream analytics.
- **Loose coupling and independent deployment:** each microservice can be developed, tested, deployed, and updated separately — for example, retraining the at-risk prediction model does not require redeploying the attendance-capture service.
- **Technology flexibility:** AI/ML components can use specialised frameworks (e.g., Python-based CV/ML libraries) while other services use different technology stacks best suited to their function.
- **Fault isolation:** a failure in one microservice (e.g., the notification service) does not bring down attendance capture or core dashboards, supporting the reliability and availability requirements identified above.
- **Extensibility:** new analytics services (e.g., a future 'classroom sentiment' module) can be added as new microservices/consumers on the event bus without redesigning the whole system.

Alternative styles were considered and rejected as the sole architecture: a purely Layered architecture would tightly couple attendance capture, analytics, and reporting logic, making independent scaling of the AI workload difficult; a pure Client-Server model would not efficiently support the continuous, high-volume sensor/video streaming required for real-time engagement analysis; and a fully Serverless approach, while attractive for burst workloads, was considered less suitable for continuously running video-stream processing. The hybrid Microservices + Event-Driven approach was judged the best fit for the assigned problem's scale, real-time, and AI-driven nature.

3. Draw the Software Architecture Diagram

The diagram below illustrates the overall structure of the proposed system, organised into seven layers: presentation, API gateway, microservices, event/message bus, AI/ML processing, data storage, and external edge devices. Arrows indicate the primary direction of data and control flow between layers.

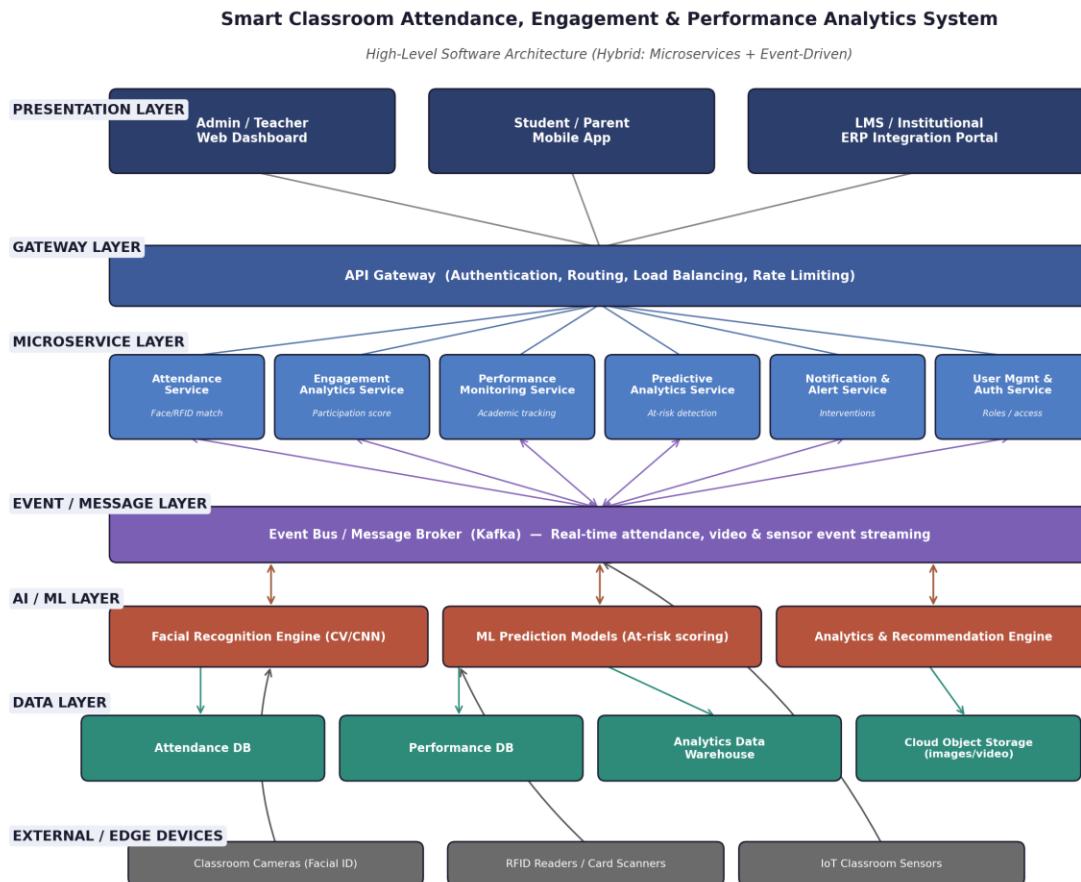


Figure 1: High-level layered/microservices architecture of the Smart Classroom Analytics System

4. Explain Components and Their Interactions

4.1 Purpose and Responsibilities of Each Component

- **Presentation Layer (Web Dashboard, Mobile App, LMS Portal):** provides role-based interfaces for teachers/admins, students/parents, and institutional systems to view attendance, engagement, and performance analytics.
- **API Gateway:** acts as the single entry point for all client requests, handling authentication, request routing, load balancing, and rate limiting before forwarding requests to the appropriate microservice.
- **Attendance Service:** receives capture events from cameras/RFID readers, coordinates identity verification, and records attendance against the correct student, class, and session.
- **Engagement Analytics Service:** computes participation and attentiveness metrics from classroom sensor/video data and publishes engagement scores.
- **Performance Monitoring Service:** aggregates assessment, assignment, and examination results to track each student's academic performance trend over time.
- **Predictive Analytics Service:** applies machine-learning models to attendance, engagement, and performance data to identify students at risk of falling behind.
- **Notification & Alert Service:** converts risk flags and system events into actionable alerts delivered to teachers, mentors, or parents.
- **User Management & Auth Service:** manages student, staff, and administrator identities, credentials, and role-based access permissions.
- **Event Bus / Message Broker:** decouples producers (cameras, sensors, services) from consumers, enabling asynchronous, real-time streaming of attendance, engagement, and sensor events.
- **Facial Recognition Engine / RFID Matching:** performs computer-vision or RFID-tag matching against enrolled student templates to verify identity.
- **ML Prediction Models:** trained models that score each student's risk level based on historical and real-time features.
- **Data Layer (Attendance DB, Performance DB, Analytics Warehouse, Cloud Storage):** persists transactional records, historical analytics, and media (images/video) used for auditing and model retraining.
- **External / Edge Devices (Cameras, RFID Readers, IoT Sensors):** capture raw attendance and classroom-activity signals at the point of instruction.

4.2 How Components Communicate

Client applications communicate with the backend exclusively through the API Gateway over secure REST/HTTPS calls. Internally, microservices publish and subscribe to events on the Kafka-based event bus for high-throughput, real-time data (attendance captures, engagement readings, sensor pings), while direct synchronous service-to-service calls are used for request/response interactions such as authentication checks. The AI/ML layer consumes streamed data from the event bus, performs inference, and writes results back to the data layer and onward to the Notification service and dashboards.

4.3 Overall System Workflow

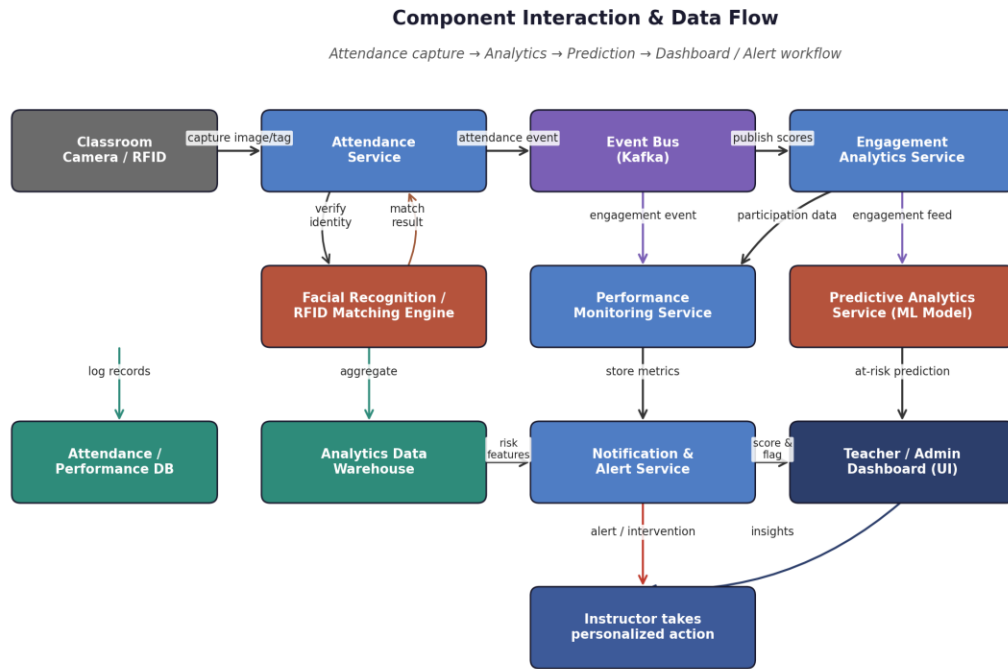


Figure 2: Component interaction and data flow — from attendance capture to instructor action

As shown in Figure 2, a classroom camera or RFID reader captures a raw identification event, which the Attendance Service forwards to the Facial Recognition / RFID Matching engine for verification. Once matched, the attendance record is published to the event bus, from which the Engagement Analytics and Performance Monitoring services consume relevant data. The Predictive Analytics Service combines attendance, engagement, and performance signals to compute an at-risk score; when a risk threshold is crossed, the Notification & Alert Service informs the instructor through the dashboard, enabling a timely, personalised intervention.

5. Discuss Quality Attributes

5.1 Scalability

The microservices are independently deployable and can be horizontally scaled behind the API Gateway and load balancer. Because facial recognition and predictive analytics are the most computationally intensive components, they can be scaled out on GPU-enabled compute instances independently of lighter services such as notifications or user management, allowing the platform to grow from a single department to an entire multi-campus institution.

5.2 Security

All client traffic passes through the API Gateway, which enforces authentication and role-based authorisation before any request reaches internal services. Biometric templates, video data, and academic records are encrypted at rest (in cloud object storage and databases) and in transit (TLS between all services). The User Management & Auth Service centralises identity and access control, and audit logging on the event bus supports traceability of every attendance and grading event.

5.3 Performance

The event-driven backbone allows attendance capture to be acknowledged to the edge device almost immediately, while heavier analytics (engagement scoring, at-risk prediction) are processed asynchronously without blocking the capture path. Caching of frequently accessed dashboard data and pre-aggregation in the Analytics Data Warehouse further reduce response times for teachers viewing real-time class summaries.

5.4 Reliability

Because services communicate through a durable message broker, if a downstream consumer (e.g., the Predictive Analytics Service) is temporarily unavailable, attendance and engagement events are retained on the bus and processed once the service recovers, preventing data loss. Each microservice can also be deployed with multiple replicas so that the failure of a single instance does not interrupt system operation.

5.5 Maintainability

The clear separation of concerns — attendance, engagement, performance, prediction, and notification each in their own service — allows individual teams to update or retrain a component (for example, improving the ML risk-prediction model) without impacting unrelated services, reducing regression risk and simplifying testing.

5.6 Availability

Redundant service instances, a resilient message broker, and cloud-based hosting with load balancing and automated failover ensure the platform remains available throughout instructional hours, which is critical since attendance capture and engagement monitoring occur continuously during every class session.

6. Justify Architectural Decisions

6.1 Rationale Behind the Architectural Choices

The combination of microservices and event-driven communication was chosen because the problem statement inherently involves multiple independent concerns (attendance capture, engagement analysis, performance tracking, and prediction) operating on continuous, high-volume streams of sensor and video data. Decomposing these concerns into separate services, connected through an asynchronous event backbone, directly satisfies the system's need for real-time processing, independent scalability of AI workloads, and resilience against partial failures.

6.2 Trade-offs Considered

- **Complexity vs. flexibility:** a microservices approach introduces additional operational complexity (service discovery, distributed monitoring, inter-service contracts) compared with a monolithic design, but this trade-off is justified by the independent scaling and deployment benefits it provides for AI-heavy workloads.
- **Eventual consistency vs. simplicity:** the event-driven design means dashboards may briefly lag behind the very latest classroom event (eventual consistency), which was accepted in exchange for higher throughput and resilience during peak attendance periods.
- **Infrastructure cost vs. performance:** running dedicated AI/ML services with GPU support increases infrastructure cost relative to a simpler rule-based system, but is necessary to achieve acceptable facial-recognition accuracy and real-time predictive insight.
- **Centralised gateway vs. single point of failure risk:** the API Gateway simplifies security and routing but must itself be deployed redundantly to avoid becoming a bottleneck or single point of failure.

6.3 Supporting Future Enhancements, System Integration, and Long-Term Maintainability

Because each capability is encapsulated as an independent microservice communicating through well-defined events and APIs, the architecture readily supports future enhancements such as additional analytics modules (e.g., sentiment or emotion analysis), integration with new institutional systems (LMS, ERP, parent-communication apps), and periodic upgrades to the underlying AI models — all without requiring a redesign of the overall system. This positions the platform to evolve alongside institutional needs while maintaining a stable, secure, and scalable foundation for accurate attendance management, improved engagement, early identification of at-risk students, and ultimately better learning outcomes.

— End of Report —